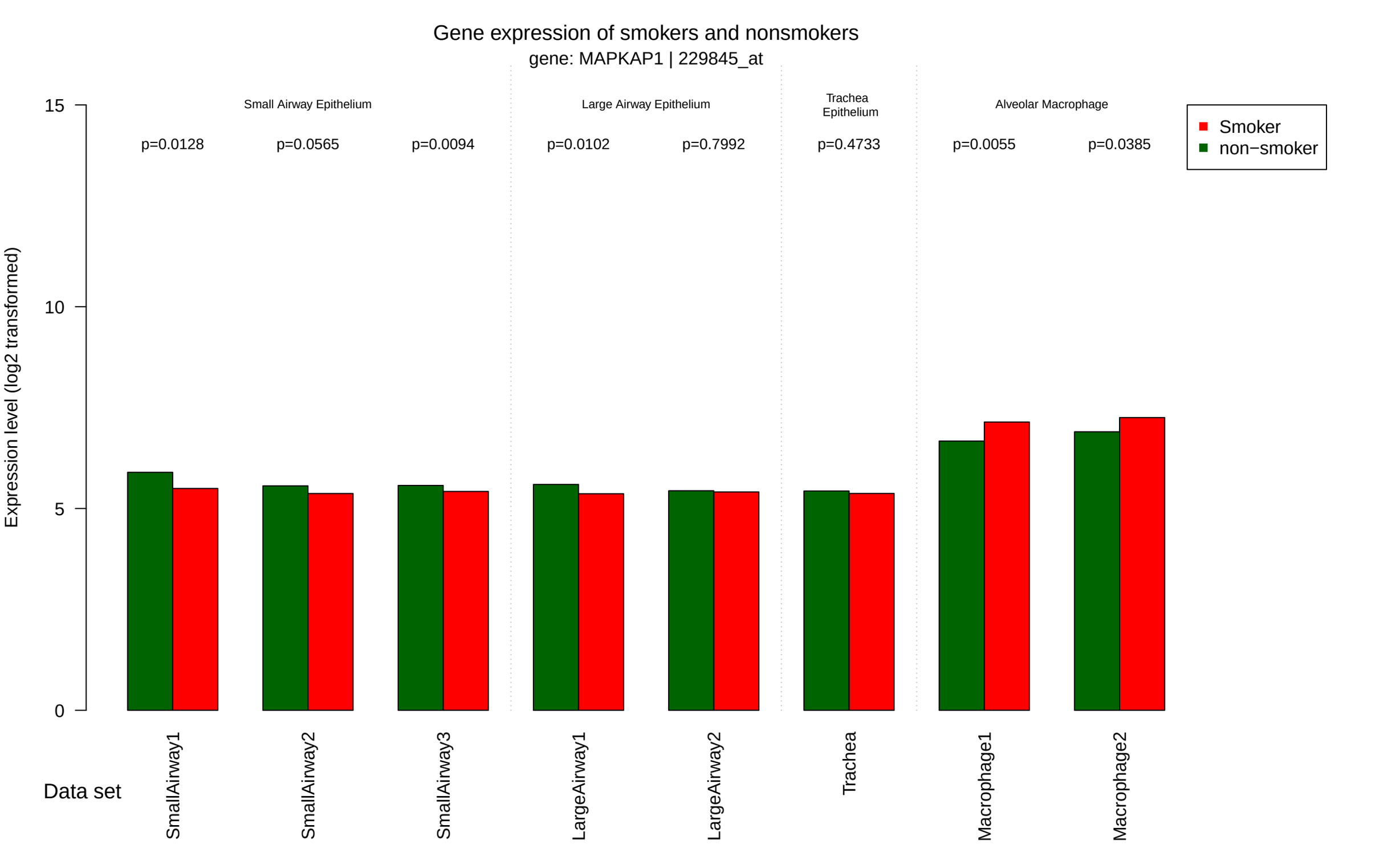
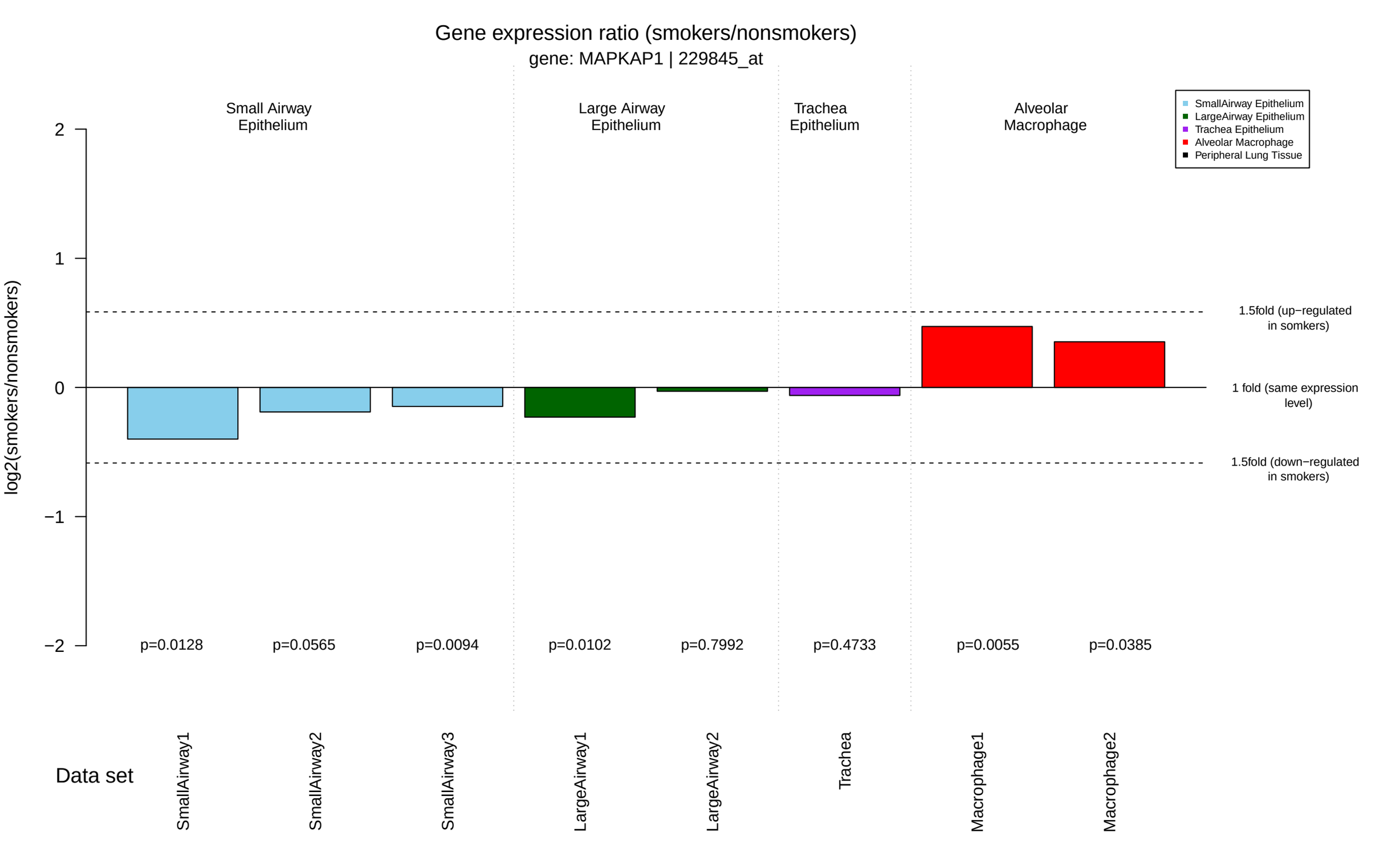
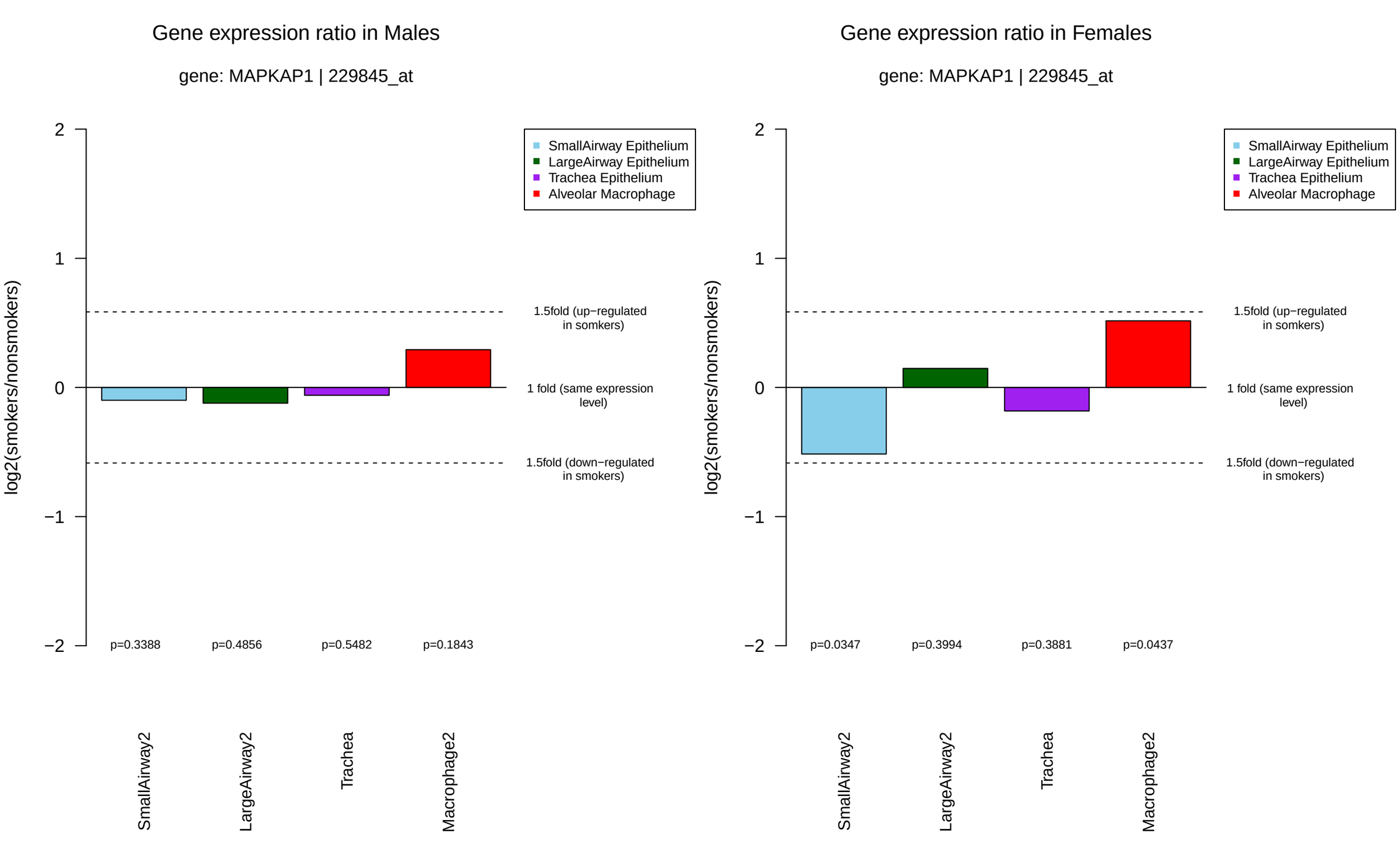
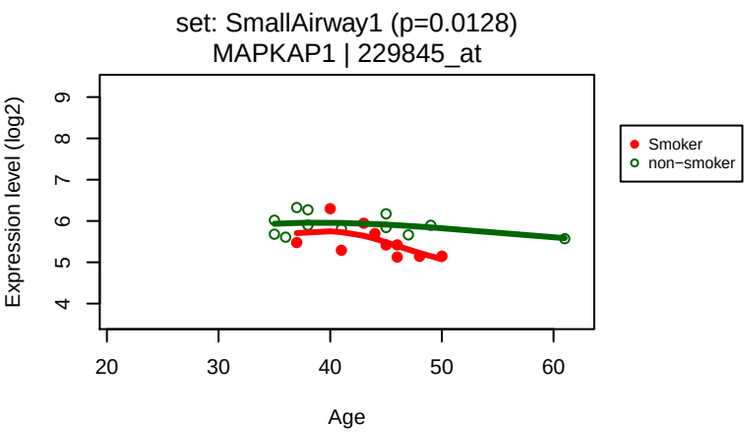
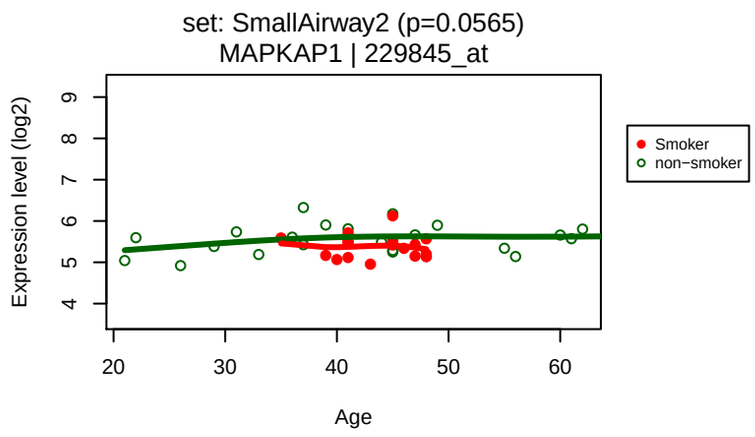
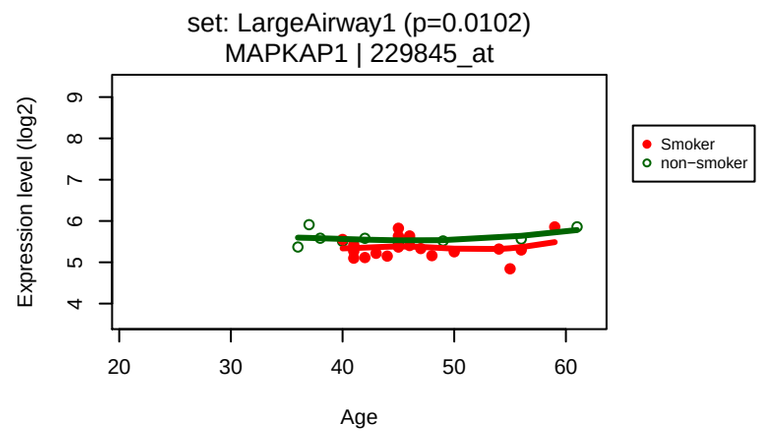

Gene expression of smokers and nonsmokers
gene: MAPKAP1 | 229845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0128
p=0.0565
p=0.0094
p=0.0102
p=0.7992
p=0.4733
p=0.0055
p=0.0385
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

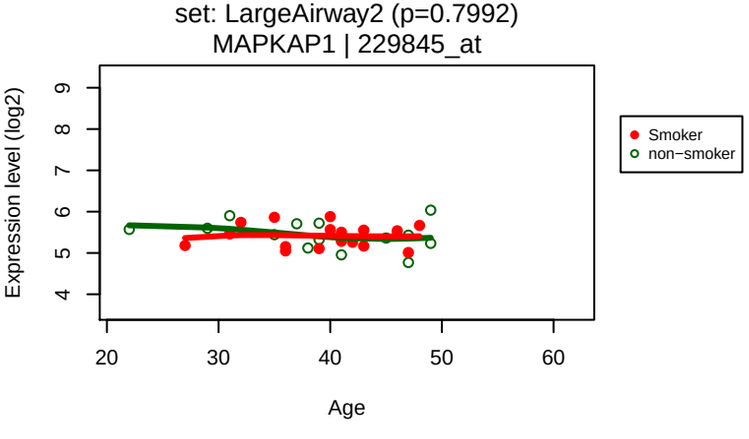

Gene expression ratio (smokers/nonsmokers)
gene: MAPKAP1 | 229845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0128
p=0.0565
p=0.0094
p=0.0102
p=0.7992
p=0.4733
p=0.0055
p=0.0385
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

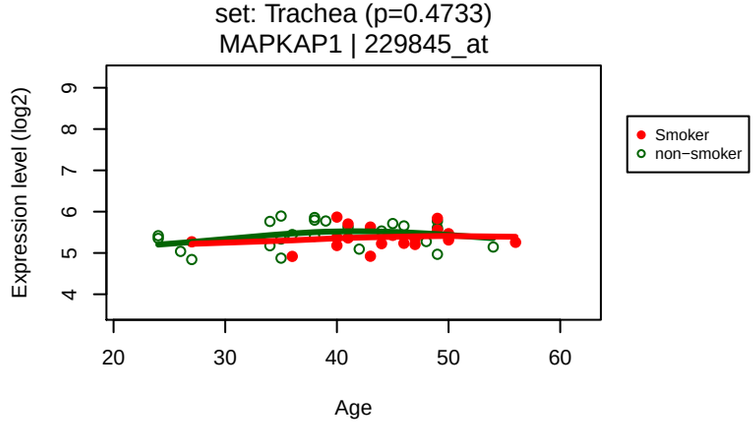

Gene expression ratio in Males
gene: MAPKAP1 | 229845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3388
p=0.4856
p=0.5482
p=0.1843
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPKAP1 | 229845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0347
p=0.3994
p=0.3881
p=0.0437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

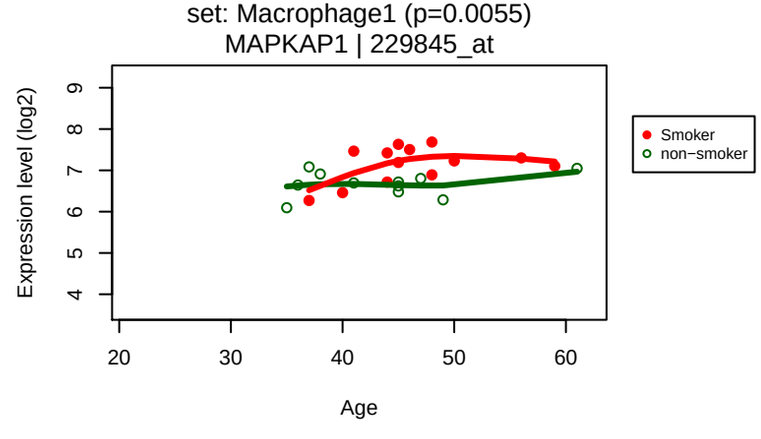

set: SmallAirway1 (p=0.0128)
MAPKAP1 | 229845_at
Expression level (log2)
Age
Smoker
non−smoker

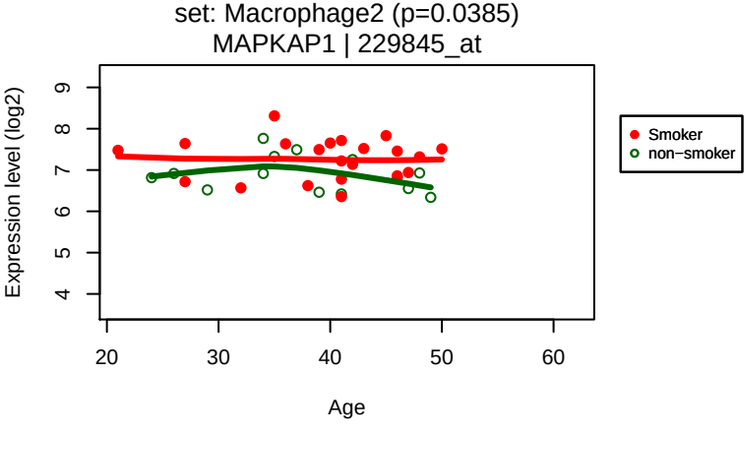

set: SmallAirway2 (p=0.0565)
MAPKAP1 | 229845_at
Expression level (log2)
Age
Smoker
non−smoker

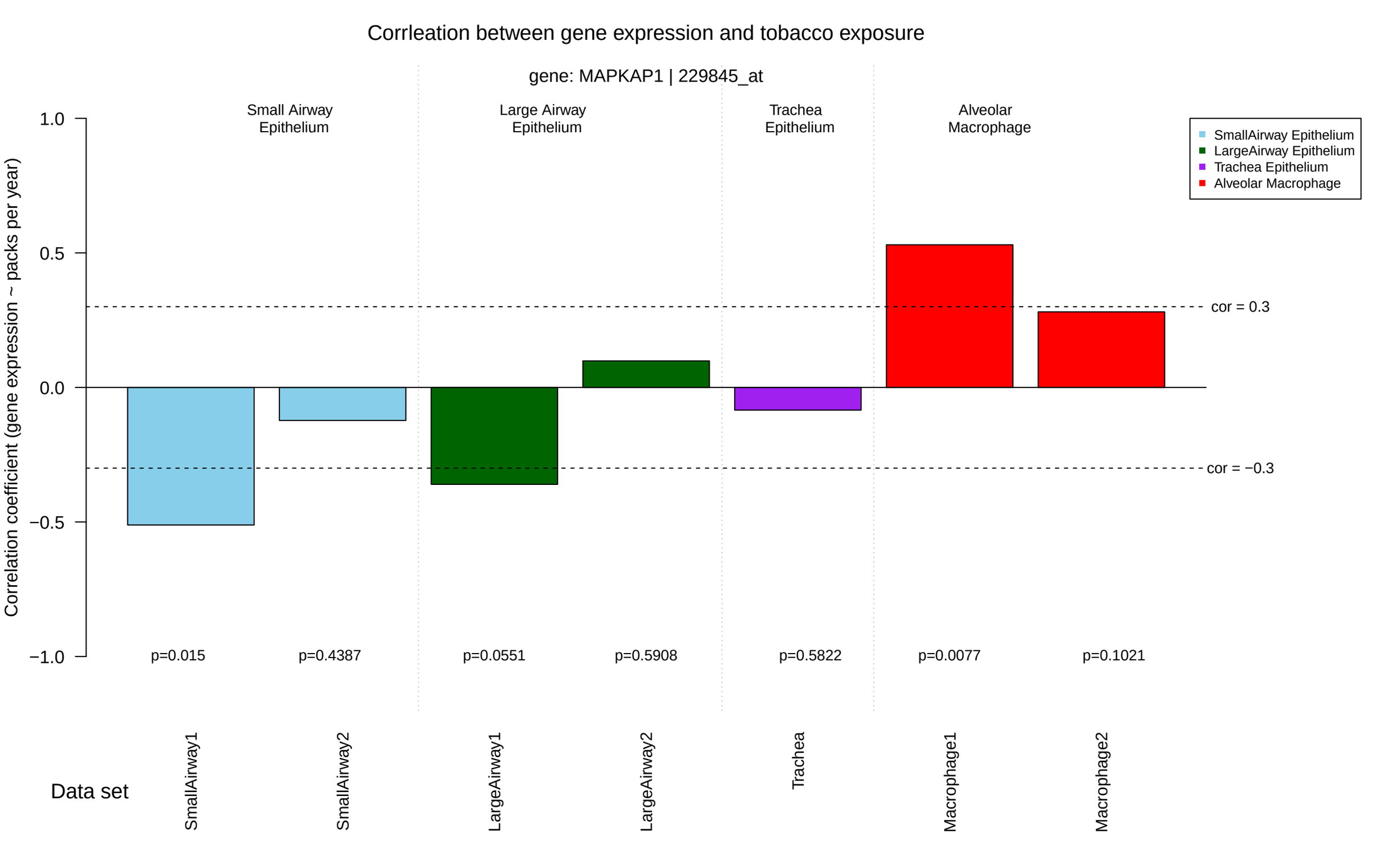

set: LargeAirway1 (p=0.0102)
MAPKAP1 | 229845_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7992)
MAPKAP1 | 229845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4733)
MAPKAP1 | 229845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0055)
MAPKAP1 | 229845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0385)
MAPKAP1 | 229845_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPKAP1 | 229845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.015
p=0.4387
p=0.0551
p=0.5908
p=0.5822
p=0.0077
p=0.1021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2